MIL-S-24521A(SH)
20 June 1986
SUPERSEDING
MIL-S-24521(SHIPS)
2 September 1975
(See 6.5)

MILITARY SPECIFICATION

# SODIUM NITRITE, TECHNICAL

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.2 Scope. This specification covers technical grade sodium nitrite for use as an anti-corrosive agent in boiler preparation.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL
PPP-D-723 - Drums, Fiber.
PPP-C-2020 - Chemicals, Liquid, Dry and Paste: Packaging of.

STANDARDS

FEDERAL
FED-STD-313 - Material Safety Data Sheet Preparation and Submission of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 5523, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.

2.1.2 Other Government documents. The following other Government documents forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF TRANSPORTATION (DOT)
Code of Federal Regulations (CFR)
49 CFR, Parts 170-179 - Hazardous Material Regulations.

DEPARTMENT OF LABOR
Occupational Safety and Health Administration (OSHA)
Code of Federal Regulations (CFR)
29 CFR, Part 1910.1200 - Hazard Communication Standard.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

(Copies of specifications, standards and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D 1193 - Standard Specification for Reagent Water. (DoD adopted)
D 1293 - Standard Test Methods for pH of Water and Waste Water. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN CHEMICAL SOCIETY (ACS)
Specifications - Reagent Chemicals.

(Application for copies should be addressed to Applied Publications, American Chemical Society, 1155 16th Street, NW, Washington, DC 20006.)



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(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 Material. Sodium nitrite shall be in the form of anhydrous crystals and shall conform to the formula, $NaNO_2$.

3.2 Chemical requirements. The sodium nitrite shall conform to the chemical requirements specified in table I when tested as specified in 4.3.

TABLE I. Chemical requirements.

| Constituent or property | Value | Percent by weight | Test paragraph |
|---|---|---|---|
| Sodium nitrite | ---- | 97.0 (minimum) | 4.3.3 |
| Sulfates (as $Na_2SO_4$) | ---- | 0.2 (maximum) | 4.3.4 |
| Chlorides (as NaCl) | ---- | 0.2 (maximum) | 4.3.5 |
| Insoluble matter | ---- | 0.5 (maximum) | 4.3.2 |
| pH value | 8 ± 1 | ---- | 4.3.6 |

3.3 Material safety data sheet. The contracting activity shall be provided a material safety data sheet (MSDS) at the time of contract award. The MSDS shall be provided in accordance with the requirements of FED-STD-313 and 29 CFR 1910.1200. When FED-STD-313 is at variance with the CFR, 29 CFR 1910.1200 shall take precedence, modify, and supplement FED-STD-313.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve



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the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Quality conformance inspection.

4.2.1 Lot and batch. For the purpose of sampling and inspection, a lot shall consist of sodium nitrite from one batch offered for delivery at one time, but not to exceed 24 consecutive hours under essentially the same manufacturing conditions and with no change in materials, providing the operation is continuous. A batch is that quantity of material which has been subjected to some unit chemical or physical mixing process intended to make the final product substantially uniform. In the event the process is a batch operation, each batch shall constitute a lot.

4.2.2 Sample for tests. Ten primary samples of equal weight, 25 grams each, shall be randomly selected from each lot of material and thoroughly mixed to form a composite sample representative of the lot. The container shall be labeled to identify the sample with the lot represented. If there are less than 10 containers in a lot, a sample shall be taken from each container. The composite sample shall be subjected to the tests of 4.3. A lot shall be unacceptable if a sample fails to meet any of the test requirements specified herein.

4.2.3 Examination of filled containers. A random sample of filled containers, taken from each lot in accordance with MIL-STD-105 at inspection level I and acceptable quality level of 2.5 percent defective, shall be examined to verify compliance with all stipulations of this specification regarding fill, closure, marking, and other requirements not involving tests. Samples shall be examined for defects of the container and closure, for evidence of leakage, and for unsatisfactory markings. Each sample filled container shall also be weighed to determine the amount of the contents. Any container in the sample having one or more defects or under required fill shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan in accordance with MIL-STD-105 the lot represented by the sample shall be rejected.

4.3 Test methods. Unless otherwise specified (see 6.2), the physical and chemical values specified in 3.1 and table I shall apply to the average of two determinations made on the unit of product.

4.3.1 Reagent water conforming to type III grade of ASTM D 1193 and reagent chemicals conforming to American Chemical Society specifications shall be used throughout the tests. Weighings shall be made to the nearest 0.0001 gram. Sodium nitrite shall be weighed as received. Unless otherwise specified herein, solutions shall be made up on a weight basis.

4.3.2 Insoluble matter. Dissolve 25 grams of sample in 250 milliliters (mL) of water, heat to boiling, and digest in a covered beaker on a steam bath for 1 hour. Filter the liquid through a tared sintered glass crucible (fine porosity with a nominal maximum pore diameter of 5 micrometers (µm) which has been dried to constant weight at 110 ± 2 degrees Celsius (°C). Quantitatively



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(d) Quantity per container if other than specified (see 3.1):

transfer insoluble matter to the crucible by thoroughly washing the beaker with hot reagent water. The combined filtrate and wash water are retained for sulfate (see 4.3.4) and chloride (see 4.3.5). Dry the filtering crucible to constant weight at 110 ± 2°C. The percent insoluble matter is calculated as follows:

$$\text{Percent insoluble matter} = \frac{100\ (A-B)}{W}$$

Where:
A = Weight of crucible plus residue in grams
B = Weight of crucible in grams
W = Weight of sample in grams

4.3.3 Sodium nitrite. Dissolve 10 grams of sample in water and dilute to 1000 mL in a volumetric flask. Add 5 mL of concentrated sulfuric acid to 300 mL of water in a 600 mL beaker. Stir, and while the solution is still warm, add 0.1N potassium permanganate until a faint pink color is produced which persists for 2 minutes. Add 40.0 mL of 0.1N potassium permanganate and stir gently. Pipet 10.0 mL of the diluted sample, slowly and with constant agitation, into the solution in the 600 mL beaker. Keep the tip of the pipet well below the surface of the liquid. Add 15.0 mL of 0.1N ferrous ammonium sulfate and allow the solution to stand for 5 minutes. Titrate the excess of ferrous ammonium sulfate with 0.1N potassium permanganate and record this value as A. Calculate the percent sodium nitrite as follows:

$$\text{Percent sodium nitrite} = \frac{0.345A}{W}$$

Where:
A = mL of 0.1N potassium permanganate used to back titrate
W = Weight of sample in aliquot in grams

4.3.4 Sulfate. Cool the filtrate and washings retained from 4.3.2. Quantitatively transfer the solution to a 500 mL volumetric flask and dilute to volume. Pipet 200 mL of this solution into a 400 mL beaker. Add methyl orange indicator. Add sufficient hydrochloric acid to make the solution just acidic to methyl orange. Add 2 mL of concentrated hydrochloric acid in excess. Heat to boiling. Add dropwise, while stirring, 10 mL of 10 percent barium chloride solution. Continue boiling for 5 minutes. Remove the solution from the hot plate and allow the precipitate to settle. Filter the solution through a tared sintered glass crucible (having a nominal maximum pore diameter of 5 µm) which has been dried to constant weight at 110 ± 2°C. Quantitatively transfer the precipitate to the filter. Wash with hot reagent water until the filtrate is free of chloride as indicated by the absence of a precipitate in a silver nitrate solution. Dry the filtering crucible to constant weight at 110 ± 2°C. The percent sodium sulfate is calculated as follows:

$$\text{Percent sodium sulfate} = \frac{60.9\ (A-B)}{W}$$

Where:
A = Weight of precipitate plus crucible in grams
B = Weight of crucible in grams
W = Weight of sample in aliquot in grams



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4.3.5 Chloride. Pipet 200 mL of sample from the solution prepared from the filtrate and washings of 4.3.2 into a 400 mL beaker. Add 5 mL of concentrated nitric acid. Boil gently for 5 minutes. Add 10 mL of approximately 0.1 silver nitrate dropwise with constant stirring and continue boiling for an additional 5 minutes. Remove solution from the hot plate. Allow the precipitate to settle. Test the supernatant liquid with a few drops of 0.1N silver nitrate for completeness of precipitation. If precipitation is not complete, add an additional 5 mL of 0.1N silver nitrate, boil as indicated above, and retest the supernatant. When precipitation is complete, filter the solution through a tared sintered glass crucible (having a nominal maximum pore diameter of 5 µm) which has been dried to constant weight at 110 ± 2°C. Quantitatively transfer the precipitate to the filter. Wash with hot reagent water. Use 1:100 nitric acid for transferring and washing of the precipitate. Dry the filtering crucible to constant weight at 110 ± 2°C. The percent sodium chloride is calculated as follows:

$$\text{Percent sodium chloride} = \frac{40.8\ (A-B)}{W}$$

Where:

A = Weight of crucible plus precipitate in grams
B = Weight of crucible in grams
W = Weight of sample in aliquot in grams

4.3.6 pH value. Dissolve 2 ± 0.005 grams of sample into 200 mL of neutral reagent water. Determine the pH value of the solution in accordance with ASTM D 1293.

4.4 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Preservation-packaging and packing levels A, B and C. Unless otherwise specified (see 6.2), sodium nitrite shall be furnished in 25-pound quantities and shall be packed in type IV fiber drums in accordance with PPP-C-2020. The drums shall conform to type II of PPP-D-723 except, when specified (see 6.2), type III drums of PPP-D-723 shall be used. Drums shall not exceed 11-1/2 inches in outside diameter by 14 inches in height. Each drum shall be provided with a plastic liner fabricated from minimum 3 mil thickness polyethylene. Seams except that of the closure shall be completely heat sealed. Liner closure shall be accomplished by twisting and securing with a vinyl plastic or embedded wire tie strip. The strip shall be a minimum of 3/16 inch in width and a minimum 4 inches in length.

5.2 Palletizing. When specified, (see 6.2) drums shall be palletized in accordance with load type IV of MIL-STD-147.



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5.3 Marking. In addition to the special marking specified in 5.3.1 and any other markings specified (see 6.2), the shipping containers shall be marked in accordance with MIL-STD-129. Each shipment shall be identified as to product, assay, net weight, name of manufacturer and brand name. Each container shall show a lot number of identification of manufacture. In addition to the yellow "OXIDIZING MATERIAL" label specified in section 173.410 of Title 49 of the Code of Federal Regulations, warning labels shall be affixed to interior packages and exterior shipping containers in accordance with 29 CFR 1910.1200.

5.3.1 Special marking. The side of each container shall be durably and legibly marked as follows:

5.3.1.1 Warning label.

"WARNING
STRONG OXIDANT
CONTACT WITH ORGANIC MATERIAL MAY CAUSE FIRE

Store in cool, dry area. Store separately from oil, grease, gasoline, paint, or any combustible material such as wood, paper, or cotton.
When not in use, keep tightly closed in original container.
Keep away from fire.
Keep away from acids.
Keep away from ammonia and ammonium salts."

5.3.1.2 Disposal instructions.

SODIUM NITRITE CAN BE AN ENVIRONMENTAL POLLUTANT.
FOLLOW APPROVED DISPOSAL INSTRUCTIONS WHEN DISCARDING
THIS MATERIAL.

6. NOTES

6.1 Intended use. The sodium nitrite covered by this specification is intended for use as a corrosion inhibitor when preparing a boiler for either wet or dry lay-up. It is also used as a passivator following boiler chemical cleaning.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Quantity required in pounds.
(c) When number of determinations for average of values differs (see 4.3).
(d) Quantity per container if other than specified (see 5.1).
(e) When type III drums of PPP-D-723 are required (see 5.1).
(f) When palletizing is required (see 5.2).
(g) Additional marking, if required (see 5.3).

6.3 Material safety data sheets. Contracting officers will identify those activities requiring copies of completed material safety data sheets prepared in accordance with FED-STD-313 and 29 CFR 1910.1200. The pertinent Government mailing addresses for submission of data are listed in appendix B of FED-STD-313.



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6.4 Subject term (key word) listing.

Chloride
Insoluble matter
Oxidant
pH
Sodium nitrite, technical

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-S-24521A(SH) | SODIUM NITRITE, TECHNICAL |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*: _____ |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
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